Q.27 Define chopper. What are different types of choppers. (CO3)

Q.28 Write the advantages of HVDC transmission. (CO5)

Q.29 Draw and explain the circuit diagram of dual converter. (CO4)

Q.30 Explain the circuit diagram of motor drive control. (CO5)

Q.31 Explain the circuit diagram of offline UPS. (CO5)

Q.32 Write down the merits and demerits of series inverter. (CO4)

Q.33 Explain the function of freewheeling diode. (CO2)

Q.34 Discuss the applications of cyclo-converter. (CO4)

Q.35 Explain Voltage source inverter. (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Define triggering of SCR. Explain different types of triggering methods. (CO2)

Q.37 Differentiable between online, offline and line interactive inverter. (CO4)

Q.38 Write short note on any two

a) V-I Characteristics of SCR

b) Constant V/F operation AC drive control

c) Fully controlled full wave bridge rectifier.

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Roll No. ....

5th Semester/ Mechatronics
Subject : Power Electronics

Time : 3 Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 An SCR is a ___ layer device (CO1)

a) 2 b) 4

c) 3 d) 5

Q.2 A TRIAC is a ___ device (CO1)

a) Unidirectional b) Bidirectional

c) Both d) None

Q.3 The process of Turn ON of an SCR is called________. (CO2)

a) Triggering b) Commutating

c) Blocking d) None of the above

Q.4 If firing angle of an SCR circuit is decreased, the output______ (CO2)

a) Remains the same b) Is increased

c) Is decreased d) None of the above

Q.5 ______ device/s can be used in a chopper circuit. (CO3)

a) BJT b) MOSFET

c) GTO d) All of the above

Q.6 A single phase full wave half controlled rectifier uses_____. (CO2)

a) 2 SCRs b) 3 SCRs

c) 4 SCRs d) 6 SCRs

Q.7 The main advantage of SMPS over linear power supply? (CO5)

a) No transformer is required

b) Higher efficiency

c) No filter required

d) Only one state of conversion

Q.8 A cyclo converter having negative half of load current is carried by______. (CO4)

a) Negative converter only

b) Positive converter only

c) Positive & Negative converter

d) Positive or negative converter

Q.9 A PWM switching scheme is used in single phase inverters to: (CO4)

a) Reduce the higher order harmonics

b) Minimize the effect of load.

c) Reduce the lower order harmonics

d) Reduce the total harmonic distortion

Q.10 UPS stands for ____. (CO5)

a) Universal power supply

b) Unintererruptive power supply

c) Universal power system

d) None of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

Q.11 SUS stands for _______ (CO1)

Q.12 Controlled Rectifier contains only diodes. (True/ False) (CO2)

Q.13 Define dv/dt triggering? (CO2)

Q.14 The duty cycle of chopper is expressed by _______ (CO3)

Q.15 Write the terminal name of UJT. (CO1)

Q.16 ______ active material used for positive plate of lead acid battery. (CO3)

Q.17 HVDC stands for _____. (CO5)

Q.18 Freewheeling diode is used to improve power factor in controlled rectifiers. (True/False) (CO2)

Q.19 The most suitable motor for control application is ________ motor. (CO5)

Q.20 Which motor is used for elevators? (CO5)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe the two transistor model of SCRs. (CO1)

Q.22 Explain Class B commutation method of SCR. (CO2)

Q.23 Explain the working principle of UJT. (CO2)

Q.24 Write any four specifications of SCR. (CO1)

Q.25 Draw the circuit diagram battery charger using thyristor. (CO5)

Q.26 Draw and explain the circuit diagram of single phase half wave controlled rectifier. (CO2)